



Fig. 2: Harry Potter's invisibility cloak is close to becoming a reality

the only way to get around it is to use cloaks fashioned out of active, electrically-active materials.

This might change in the future with more advanced passive meta-materials, but for now, active designs are the way forward. Research into active invisibility cloaks is currently being carried out by multiple research groups all over the world, but none have yet been built.

Invisibility cloaks are supposed to hide things by bending light around these that would not allow any to be reflected back at a potential viewer, which would allow the invisible thing to be seen.

Invisibility cloaks, which are very successful in movies and are so close to being successful in labs, have a crucial weakness. These make things invisible to the perception of the audience their creators had in mind.

Previous work on cloaking objects from view has relied on so-called meta-materials made of artificial materials with unusual properties. The composite structures used for cloaking cause light beams to bend around an object and then meet on the other side, resuming their original path, making the ob-

ject appear invisible.

The great unappreciated weakness of invisibility cloaks is that these only make things invisible to human eyes. A cloak made to hide things from humans would be able to bend all colours of visible light, but might not be able to do the same with waves of heat or sound.

It has never been clear whether it was even theoretically possible to make an invisibility cloak that could hide the same object from light, heat and sound. The problem is not just that the cloak material has to bend light, but that it has to have an effect on the light, powerful enough to change its behaviour in very specific ways.

A mirror can change the behaviour direction of a beam of light dramatically, and can hide a person standing behind it, for example, but does not do those things with enough subtlety to convince anyone that it is not there. An advanced meta-material from the category of meta-materials often used in attempts at invisibility might have a powerful, innate ability to scatter light, but could melt when exposed to a lot of heat, or catch fire when in contact with electricity.

Researchers have addressed the need for a material that could manipulate more than one type of material by making a cloak out of two different layers of material with different properties, but each has to be able to shield the other from view within its own slice of the energy spectrum.

A team of researchers has built a double layer of material as a cloak that can keep objects from being seen using either heat or electricity as a viewing medium, using a layer of silicon that attracts and concentrates both electrical current and heat flow, and an inner layer that is actually an empty cavity that scatters both current and heat away from itself. So when an object is placed within the invisibility zone, heat and electricity are diverted around it and scattered around the inside of the cavity. The outer shell then pulls both heat and electricity towards itself, removing the means of being seen from the neighbourhood of the object being hidden.

Holes are drilled in the outer layer and filled with a material called polydimethylsiloxane, which conducts heat and electricity little or not at all compared to silicon. The combination of two materials scatters heat and electricity randomly enough to make the accumulated energy difficult to notice.

Invisibility to electronics

The combined effect of invisibility on heat and electricity could be used to help eliminate unwanted electrical and heat energy from sensitive electronic components, which could eliminate much of the source of static that make radio reception less clear or sap the performance of the devices themselves. The technique, which boils down to protecting an object from heat and electricity by scattering it away, might also be useful in overcoming the performance barriers that have made it difficult to increase the efficiency of